

“Design and Development of EstateX – Real Estate Listing & Virtual Tour Portal”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
&
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



Declaration

The Project Report entitled “**Design and Development of EstateX – Real Estate Listing & Virtual Tour Portal**” is a record of bonafide work done and submitted by **SAI KRISHNA – 2520090156, NAVANEETH – 2520090218, CHAITHANYA – 2520090099** in partial fulfillment of the requirements for the award of the degree of B. Tech in Computer Science and Information Technology to K L University.

The report presents the analysis, design, implementation and testing of EstateX, a web-based real-estate listing and virtual-tour portal. The work has been prepared for academic evaluation and reflects the project implementation carried out by the team.

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CERTIFICATE

This is to certify that the mini project based report entitled “**Design and Development of EstateX – Real Estate Listing & Virtual Tour Portal**” is a bonafide work done and submitted by **SAI KRISHNA – 2520090156**, **NAVANEETH – 2520090218**, **CHAITHANYA – 2520090099** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Computer Science and Information Technology, K L (Deemed to be) University.

The report documents the real-estate portal architecture, database design, implementation modules, API integration, authentication, testing and deployment approach developed as part of the academic project.

Signature of the Course Coordinator

Signature of the Guide

Signature of the HOD

ACKNOWLEDGEMENT

The successful completion of this project would not have been possible without the timely support, guidance and encouragement received from the faculty members, guide and institution. We express our sincere gratitude to our project guide, **Gayathri**, for providing direction, feedback and continuous encouragement during the development of EstateX.

We thank the Department of Computer Science and Information Technology, KLU University, for providing the academic environment, laboratory facilities and resources required to plan, develop and document this project. The project provided an opportunity to apply concepts from web development, database systems, software engineering and backend development to a practical problem.

We also thank our teammates, classmates and all those who provided suggestions during the design, implementation and review stages. Their feedback helped us improve the usability, structure and technical clarity of the EstateX application.

Abstract

EstateX is a web-based real estate listing and virtual-tour portal designed to simplify the process of discovering, comparing and enquiring about properties. The project addresses the fragmentation that buyers often experience when property information is distributed across multiple portals, social-media posts, agent contacts and informal listings. EstateX provides a single digital interface where users can search properties by location, listing type and property type, review important property details, and initiate enquiries with agents.

The proposed system is structured as a full-stack application. The frontend is developed using React.js with a Vite-based development environment and responsive UI components. A Flask REST API provides the backend services for authentication, property management, agents, enquiries, favourites and administrative functions. MongoDB is used as the application database for storing flexible property, user and enquiry records. Environment variables are used for configuration, while CORS and token-based authentication support controlled communication between the frontend and backend.

EstateX is initially focused on Hyderabad and is designed around two major stakeholder groups: property buyers/tenants and agents. Search and filtering improve property discovery, while property listing and enquiry modules support the agent side of the platform. The system also includes an EMI calculator and is designed to support location-based discovery and 360-degree virtual property experiences. The architecture is modular so that additional features such as advanced verification, payments, notifications, analytics and AI-assisted recommendations can be introduced later.

The project demonstrates how frontend engineering, REST API development, database design, authentication, user experience design and software testing can be combined to build a practical real-world digital platform. EstateX is therefore both an academic full-stack project and a foundation for a future real-estate technology product.

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Chapter sequence: The report retains the structure of the supplied KLH project-documentation template. The content has been adapted to EstateX without changing the chapter order or academic reporting pattern.

Project identity: EstateX — Real Estate Listing & Virtual Tour Portal. Initial target geography: Hyderabad. Primary users: property buyers/tenants and real-estate agents.

Core modules: Authentication and user management; property listings and verification; search and filters; property detail pages; agent dashboard; lead and enquiry management; favourites; EMI calculator; administration; location/virtual-tour support.

Technology summary: React.js + Vite frontend, Flask REST API backend, MongoDB database, PyMongo, Flask-CORS, python-dotenv, token-based authentication, Git/GitHub, VS Code and Postman.

CHAPTER -1 INTRODUCTION

1.1 Background of the Project

EstateX is conceived as a digital real-estate portal that brings property discovery, listing management and buyer-agent communication into a single web application. Property seekers frequently encounter listings across multiple channels, and important details such as price, area, location, amenities and agent contact information are often presented in inconsistent formats. EstateX establishes a structured interface in which these details can be represented as searchable property records.

The initial market focus is Hyderabad, where buyers and tenants commonly search by locality, budget, property type and configuration. The system therefore emphasizes practical search inputs such as Buy or Rent, location, property type and price-related information. Property cards and detail views are designed to present the information required for an initial decision without forcing users to move between several pages or external sources.

From a software-engineering perspective, EstateX provides a practical case study in full-stack development. The frontend manages interactive user experiences, the Flask backend exposes REST APIs and business logic, and the database stores users, properties, agents, enquiries and related records. This separation of responsibilities makes the project easier to maintain and extend.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

An important implementation consideration is consistency between the user interface and the stored data. EstateX treats the backend API as the contract between these layers, so field names, data types, authentication requirements and error responses should remain predictable. This approach makes debugging easier because a problem can be isolated to the interface, API or database rather than being investigated as one undivided system.

The design also considers maintainability. New modules should be added through separate React components, Flask blueprints and database collections or fields where appropriate. This prevents unrelated features from becoming tightly coupled. Clear naming, reusable functions, environment configuration and Git-based version control help the team continue development without losing the working structure already established.

For project demonstration, the most important evidence is the complete user flow. A reviewer should be able to see property search, listing details, authentication, agent listing management, enquiry submission and the supporting utility features. API requests and database records can then be shown separately to demonstrate that the visible interface is backed by an actual full-stack implementation rather than static screens.

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1.2 Problem Statement

The real-estate discovery process is often fragmented. A user may find a property through one platform, obtain the agent's phone number from another source, calculate affordability separately, and then manually track enquiries or shortlisted properties. This creates unnecessary effort and makes it difficult to maintain a consistent view of available information.

EstateX addresses the problem of fragmented property discovery by providing a unified listing portal with structured search, property details, agent information and enquiry workflows. Instead of treating a listing as only an image and a phone number, the system represents it as a data record with searchable attributes such as listing type, property type, location, bedrooms, bathrooms, area, price and amenities.

The technical problem is to provide this experience through a reliable full-stack architecture. The frontend must communicate with backend APIs, the backend must validate requests and manage records, and the database must store information in a way that supports fast retrieval and future expansion. Authentication and role-based access are also required so that buyers and agents receive appropriate capabilities.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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1.3 Scope of the Project

The scope of EstateX covers a responsive web application for property buyers, tenants and agents. Users can browse available properties, search using relevant filters, open detailed property views, save favourites where enabled, and submit enquiries. Agents can create and manage property listings through authenticated workflows.

The project also covers database-backed storage, REST API communication, authentication, agent-oriented property management and an EMI calculator. Location-based discovery and virtual-tour presentation are part of the project vision and can be connected to map and 360-degree viewing services as the corresponding modules are completed.

The current academic scope is focused on demonstrating the core workflow rather than implementing every production feature of a commercial property marketplace. Online payments, large-scale moderation, advanced recommendation models, enterprise analytics and full production hosting are considered extension areas rather than prerequisites for the academic prototype.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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1.4 Objective(s)

The primary objective of EstateX is to design and develop a user-friendly real-estate portal that reduces the effort required to discover and enquire about properties. The application should present property information in a consistent format and allow users to narrow results using practical search filters.

The technical objective is to build a modular full-stack application using React.js on the frontend, Flask on the backend and MongoDB for persistence. REST APIs should provide clear interfaces for authentication, properties, agents, enquiries, favourites and administrative operations.

A further objective is to demonstrate database design and software-engineering practices expected in an academic project: validation, authentication, CRUD operations, modular routes, environment configuration, testing, version control and documentation.

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1.5 Importance of the Application / 1.6 Target Users

EstateX is important because property decisions depend on the availability and clarity of information. A structured portal can reduce repeated manual searching and make it easier to compare properties using common attributes. For agents, a centralized listing workflow can reduce the effort of repeatedly sharing the same property information with prospective customers.

The primary users are property buyers and tenants who need searchable listings and a convenient enquiry mechanism. A second major user group is real-estate agents who create listings, update property information and respond to customer leads. Administrative users may also be introduced to verify listings and monitor platform activity.

The project also has academic importance because it combines frontend engineering, backend API design, database management, authentication and deployment concepts in one application. It demonstrates how classroom concepts can be translated into a practical product architecture.

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CHAPTER -2 SYSTEM REQUIREMENTS

2.1 Hardware Requirements

EstateX can be developed on a standard student development laptop or desktop. A modern multi-core processor, 8 GB or more RAM, sufficient SSD storage and a stable internet connection are adequate for frontend and backend development, local database operation and API testing. A larger memory configuration can improve the experience when the IDE, browser, database tools and development servers are running simultaneously.

For demonstration, the application can be accessed from a desktop, laptop, tablet or smartphone with a modern web browser. The user device does not need a specialized graphics card because the core portal is a web application. Higher graphics capability can become useful when rich 360-degree property experiences are introduced.

For a future production deployment, the backend and database can be hosted on cloud infrastructure with resources scaled according to traffic. The academic version can run locally using the Vite development server, Flask server and local MongoDB service.

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2.2 Software Requirements

The development environment requires a current operating system, Visual Studio Code or another code editor, Node.js and npm for the React frontend, Python for the Flask backend, and MongoDB for data persistence. A modern browser such as Chrome, Edge or Firefox is required for UI testing.

The backend uses Flask and supporting Python packages such as Flask-CORS, PyMongo and python-dotenv. Authentication utilities and secure password handling are used as appropriate to the implementation. The frontend uses React.js and Vite, with JavaScript and CSS for application behavior and presentation.

API testing can be performed with Postman, while Git and GitHub provide version control and collaboration. Map and virtual-tour capabilities can use external services when their integration is enabled. Environment variables keep configuration such as database connection strings and secret keys outside the source code.

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2.3 Development Tools and Frameworks

React.js provides the component-based frontend architecture. Vite provides a fast development server and build workflow. The project separates reusable UI components such as authentication forms, property cards, property details, listing forms and calculators so that changes can be made without rewriting the entire page.

Flask provides the backend application layer. Flask blueprints allow modules such as authentication, properties, agents, enquiries, favourites and administration to remain logically separated. This structure supports maintainability and makes it easier to test individual API groups.

MongoDB is used for flexible document storage. Git is used to track source-code changes and GitHub provides a remote repository. Postman is useful for testing REST endpoints, while VS Code provides the main development environment. These tools together support an iterative workflow from coding to testing and documentation.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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CHAPTER -3 TECHNOLOGY STACK

3.1 Front-End Design

The EstateX frontend is implemented using React.js and a Vite development setup. React components divide the interface into reusable units, allowing the navigation bar, search controls, property cards, modal dialogs, authentication forms and calculators to be maintained independently.

The user interface follows a responsive approach so that the main property-discovery workflow can adapt to different screen sizes. State is used to control selected filters, current user information, property collections, modal visibility and API status. Form inputs are validated before data is sent to the backend.

The design emphasizes clear property information. A listing can display its title, location, price, property type, listing type, bedrooms, area, images and agent information. This structured presentation helps users understand a property quickly and provides a consistent foundation for future map and virtual-tour experiences.

For future growth, the implementation should avoid hard-coding assumptions that only work for a small demonstration. Validation, modular routes and environment-based configuration make it easier to move from local development toward a hosted application without rewriting the core product.

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The backend is implemented using Flask and exposes REST-style API routes to the React application. The application factory pattern creates the Flask app, loads configuration and initializes the database connection before registering modular blueprints.

Property routes support listing retrieval, filtering and authenticated create, update and delete operations. Additional blueprints are organized for authentication, agents, enquiries, favourites and administration. This modular design keeps business responsibilities separated and makes the API easier to extend.

Flask-CORS is configured so that the frontend development server can communicate with the backend. Environment variables provide database and secret configuration. The backend also performs validation and authorization checks so that protected operations are not treated as public requests.

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3.3 Database

MongoDB is the primary application database for the current EstateX architecture. Its document-oriented model is suitable for property records because listings can contain flexible fields such as amenities, images, agent information and optional virtual-tour data.

Core collections can include users, properties, agents, enquiries, favourites and administrative records. A property document can store values such as title, location, city, propertyType, listingType, price, bedrooms, bathrooms, areaSqft, amenities, images and agent references. User and agent records support ownership and contact workflows.

Indexes can be added to frequently searched fields such as location, listing type and property type. The backend should validate identifiers and input values before database operations. The design also leaves room for future collections such as verification records, notifications, saved searches and analytics events.

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3.4 Version Control

Git is used as the version-control system for EstateX. Version control provides a history of changes and makes it possible to recover earlier working versions when a new change introduces an error.

The project repository is maintained on GitHub. A clear repository structure separates frontend and backend code and avoids committing generated dependencies, environment secrets and local database files. The .gitignore configuration is therefore an important part of the project.

For team development, changes can be organized into meaningful commits and branches. Before pushing code, the team should verify that private configuration values are not included. GitHub also provides a useful source for project demonstration and academic submission.

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3.5 APIs and External Services

The EstateX frontend communicates with the Flask backend through HTTP requests. The API layer acts as the controlled interface between the user interface and persistent data. Endpoints are organized around resources such as authentication, properties, agents and enquiries.

Location functionality can be supported using a mapping service such as Mapbox when the map module is enabled. A 360-degree viewer can present immersive property media when compatible panoramic assets are available. These services are treated as integrations rather than hard-coded application logic.

Future external integrations may include email or SMS notifications, payment services, identity verification, cloud image storage and analytics. Each integration should be isolated behind a service layer so that the core EstateX modules remain maintainable if a provider changes.

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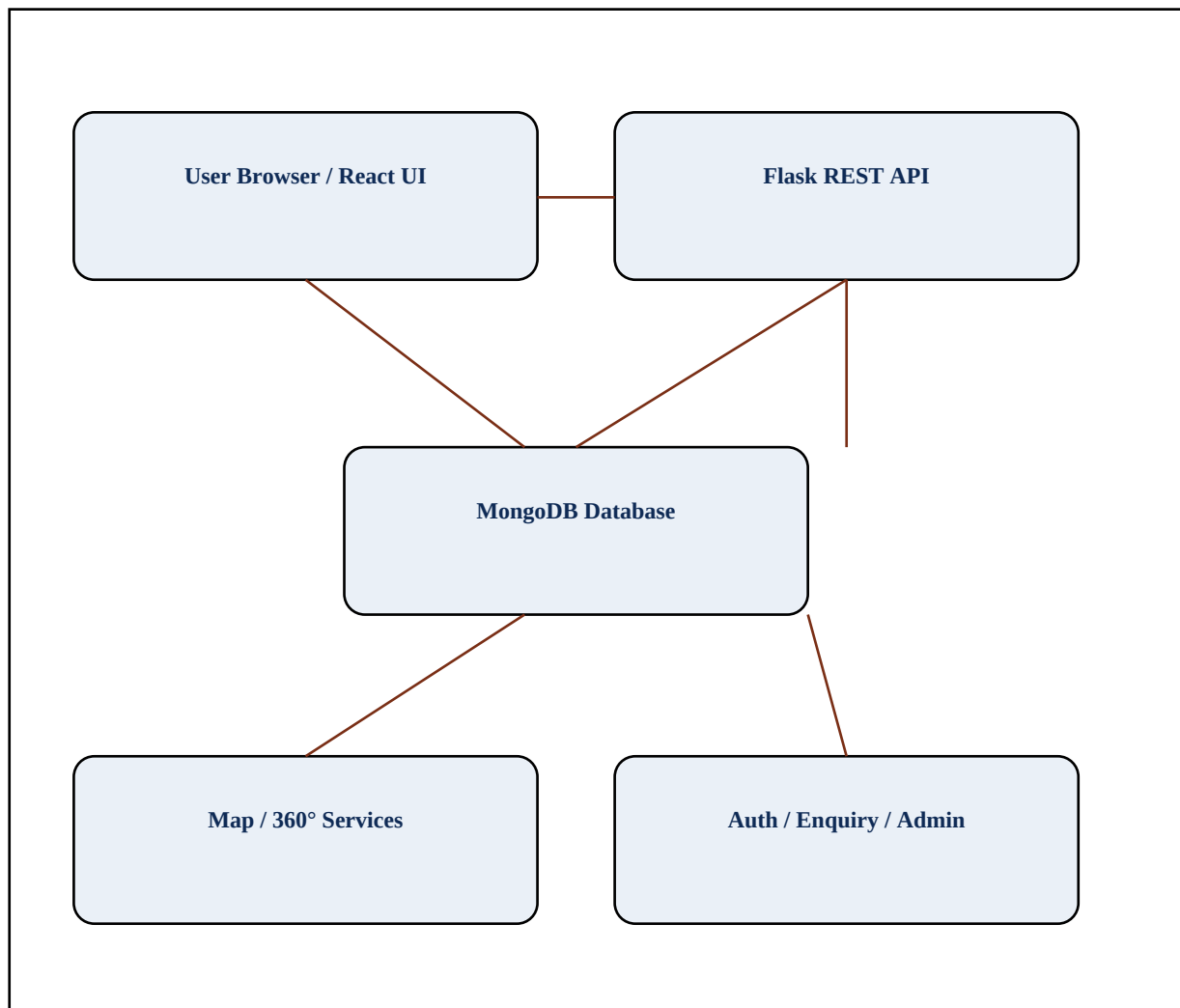
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CHAPTER -4 SYSTEM ARCHITECTURE

4.1 High-level architecture diagram



EstateX layered architecture: presentation, API/business services, persistence and integrations.

4.2 Frontend Layer

The frontend layer is responsible for presenting the EstateX interface and handling user interaction. React components manage navigation, search controls, property cards, property-detail dialogs, authentication modals, listing forms and the EMI calculator.

Frontend state represents information that changes during a session, including the selected search mode, location, property type, property results, current user and modal state. API responses are converted into UI-friendly data and rendered without requiring a full browser reload.

Responsive CSS controls the visual presentation across desktop and mobile layouts. The design is intentionally component-based so that new features such as saved searches, comparison views or richer virtual-tour controls can be introduced without redesigning the entire application.

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4.2 Backend Layer

The Flask backend is the service layer of EstateX. It receives HTTP requests, validates input, checks authentication and authorization, performs database operations and returns JSON responses to the frontend.

Blueprints provide module separation. Authentication handles registration and login, properties handles property CRUD and filters, agents handles agent-related data, enquiries records buyer interest, favourites manages saved properties and administration provides controlled management functions.

The backend also acts as the security boundary. Sensitive configuration is loaded from environment variables, protected endpoints require authenticated requests, and input validation reduces the risk of malformed data reaching the database.

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4.3 Deployment Architecture

During development, EstateX can run as three local services: the React/Vite frontend on port 5173, the Flask API on port 5000 and MongoDB on port 27017. The browser communicates with the frontend, and the frontend communicates with the Flask API.

For production, the frontend can be deployed as a static web application while the Flask backend can run on a managed server or container platform. A managed MongoDB deployment can provide remote persistence, backups and operational monitoring. Exact hosting providers can be selected according to cost and project requirements.

Environment-specific values such as API base URLs, database URIs and secret keys should be configured outside source code. HTTPS should be enabled in production and CORS should be restricted to trusted frontend origins rather than using broad development settings.

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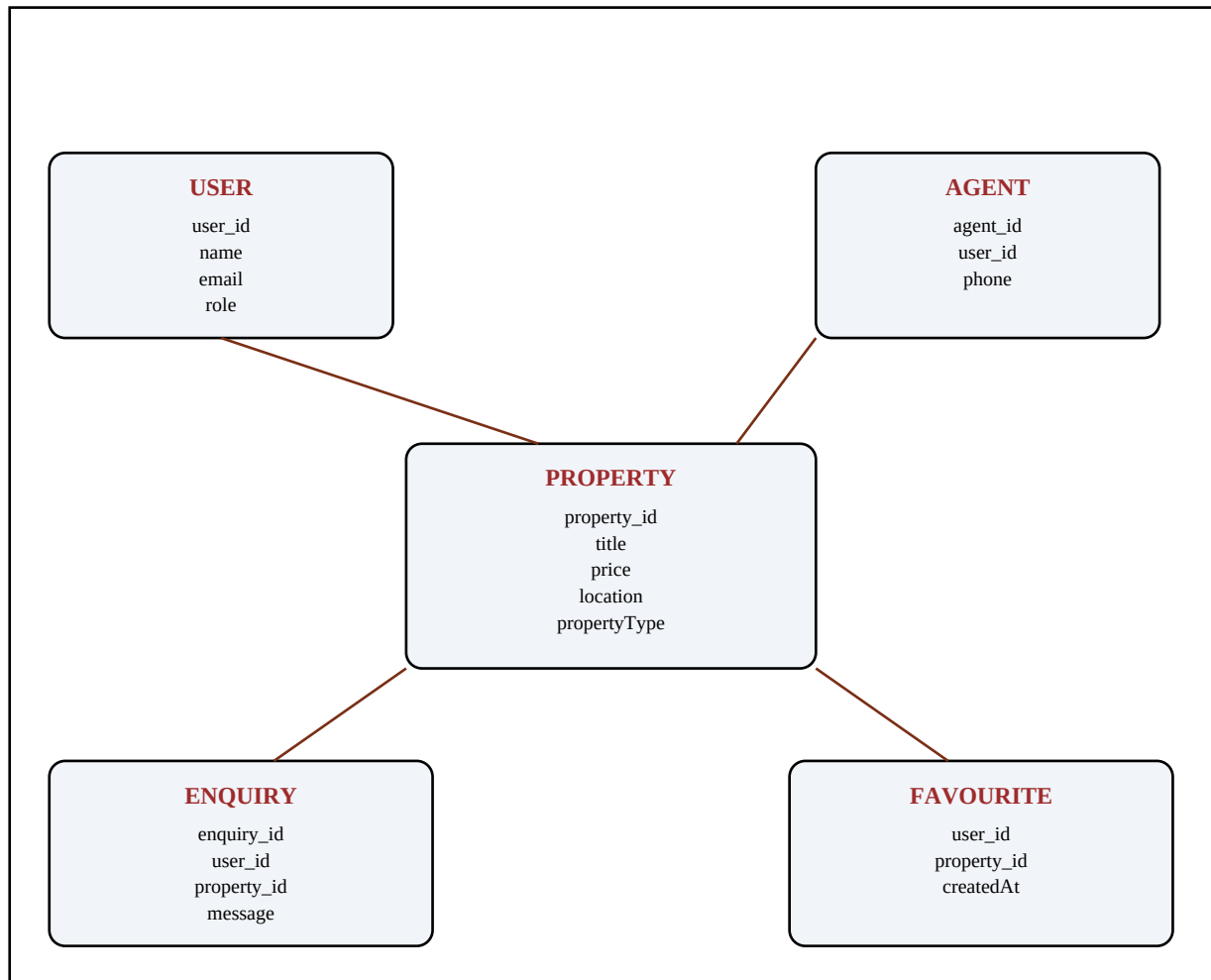
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CHAPTER -5 DESIGN

5.2 ER Diagram / Database Schema



Simplified EstateX entity relationships. Additional fields may be introduced as implementation evolves.

CHAPTER -6 IMPLEMENTATION

6.1 Module-Wise Implementation

EstateX is implemented as a collection of modules rather than one large application component. Authentication provides registration and login, the property module manages listing records, the agent module manages agent-related operations, and the enquiry module records buyer interest.

The favourites module allows a user to retain properties for later review. Administrative functions provide a controlled place for future verification and moderation. The frontend presents these modules through reusable React components and modal workflows.

This modular structure reduces coupling. For example, changes to property filtering can be implemented within the property API and corresponding frontend search state without changing authentication logic. The approach also makes it easier to test modules independently.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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6.2 Front-End Logic

The React application maintains state for the current user, property collection, search inputs, active property category, modal visibility and backend status. When the user changes a search filter, the frontend constructs an API request with the relevant parameters and updates the property list using the response.

Reusable components reduce duplicate markup. Property cards display summary information, while a property modal can present richer details. Authentication and listing forms use controlled inputs so that their values can be validated and submitted in a predictable manner.

The frontend also handles user feedback. Loading indicators, error messages and success notifications communicate the status of requests. This is important for operations such as login, publishing a listing or sending an enquiry because users need to know whether an action reached the backend.

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6.3 API Integration

API integration connects the React interface to the Flask service. A centralized API utility can expose functions such as `checkHealth`, `getProperties`, `createProperty`, `login`, `register` and `seed` or initialization operations. Centralizing requests avoids duplicating URL and header logic across components.

GET requests retrieve property collections and individual records, while POST, PUT and DELETE requests support protected management operations. Authentication tokens can be attached to protected requests so that the backend can identify the current user.

The API contract should remain predictable: successful requests return JSON data with appropriate status codes, while validation and authorization errors return informative error responses. This allows the frontend to display meaningful messages instead of exposing raw server errors.

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Authentication establishes the identity of a user, while authorization determines which operations that user may perform. EstateX distinguishes between general users and agent-oriented operations so that property creation and management are not exposed as unrestricted public actions.

Passwords should never be stored as plain text. The backend should store secure password hashes and issue a signed token or equivalent session credential after successful authentication. The frontend stores only the required session information and removes it during logout.

Protected API routes validate the supplied credential before executing sensitive operations. Role checks can then restrict actions such as creating, editing or deleting property listings. This layered approach improves security and also provides a foundation for future administrator verification workflows.

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CHAPTER -7 FEATURES

7.1 List of Main Features

The principal EstateX features are property discovery, location and category filtering, detailed property views, agent listing management, user authentication, enquiries and lead capture, favourites, an EMI calculator and administrative controls. The project also provides a foundation for map-based discovery and 360-degree virtual property experiences.

Property discovery is the central workflow. Users can select a transaction mode such as Sale or Rent and refine results using location and property type. Each result presents important information so that users can decide which listings deserve closer inspection.

Agent functionality supports the supply side of the marketplace. Authenticated agents can publish and maintain property information, allowing the platform to keep listings structured rather than relying on manually edited promotional pages.

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7.2 Property Discovery

When a user opens the EstateX search area, the interface collects the selected transaction type, location and property category. These values are passed to the property API as query parameters. The backend applies the relevant filters and returns matching property documents.

The frontend renders the returned records as cards or list items. A property can then be opened in a detail view showing images, pricing, location, configuration, area, amenities and agent contact information. This keeps the discovery workflow compact while still allowing detailed inspection.

Future search improvements can include price ranges, number of bedrooms, area ranges, map boundaries and sorting. These enhancements can be added to the same API pattern without changing the overall architecture.

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7.2 Property Listing and Agent Workflow

An agent begins by authenticating through the EstateX login workflow. Once authorized, the agent can open the property-listing form and enter information such as title, location, price, listing type, property type, bedrooms, bathrooms, area, amenities and images.

The frontend sends the listing data to the Flask property endpoint. The backend validates the request, identifies the authenticated agent and stores the property document in MongoDB. The newly created listing can then appear in the discovery results.

Update and delete operations follow the same authorization model. The agent can correct outdated details or remove a listing that is no longer available. Future verification workflows can add a status field so that administrators can approve or reject listings before public display.

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7.2 Enquiry and Lead Management

An enquiry connects a prospective buyer or tenant with a property and its agent. The user can submit interest after reviewing a listing, providing contact information and an optional message or requirement.

The backend records the enquiry as a structured document linked to the user and property. This allows the agent workflow to retrieve leads associated with its listings instead of relying on informal messages alone.

The same design can support future lead states such as New, Contacted, Scheduled, Converted or Closed. Notifications can also be introduced so that agents receive timely updates when a new enquiry is created.

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7.2 EMI Calculator and Favourites

The EMI calculator provides an affordability-oriented utility within the EstateX interface. Users can enter the principal amount, interest rate and loan duration to estimate a periodic repayment. The calculator is independent from property persistence, so it can be used before or after viewing a listing.

Favourites provide a lightweight way for users to retain interesting properties. A favourite record can connect a user with a property identifier. The user can later retrieve saved properties and compare them without repeating the original search.

These features improve the continuity of the property-discovery workflow. Future versions can add side-by-side comparison, affordability indicators and saved-search alerts while preserving the same user-centered design.

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7.2 Technical Working

Technically, each feature is implemented through a combination of React state, API calls and backend persistence. The frontend manages interaction while the Flask API remains responsible for validation, authorization and database operations.

The system uses JSON as the common data-exchange format. A property returned by the backend can contain fields such as title, location, city, propertyType, listingType, price, bedrooms, areaSqft, primaryImage and amenities. Keeping a consistent field contract reduces frontend complexity.

The modular approach also improves debugging. If a property card displays incorrect data, the team can inspect the API response separately from the React rendering logic. If the API returns an error, the backend route and database operation can be tested independently in Postman.

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CHAPTER -8 TESTING

8.1 Postman Testing

Postman is used to test EstateX REST endpoints independently of the frontend. This is useful because an API can be validated before UI integration, allowing the team to determine whether an issue belongs to the backend or the browser application.

Authentication endpoints can be tested with valid and invalid registration or login payloads. Property endpoints can be tested with GET requests for listing retrieval and POST, PUT and DELETE requests for authenticated agent operations. The response status code and JSON body are checked against the expected contract.

Postman collections can group requests by module. Environment variables can store the API base URL and temporary authentication values, making it easier to switch between local and deployed environments without editing every request.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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8.1 Postman Testing

Postman is used to test EstateX REST endpoints independently of the frontend. This is useful because an API can be validated before UI integration, allowing the team to determine whether an issue belongs to the backend or the browser application.

Authentication endpoints can be tested with valid and invalid registration or login payloads. Property endpoints can be tested with GET requests for listing retrieval and POST, PUT and DELETE requests for authenticated agent operations. The response status code and JSON body are checked against the expected contract.

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8.2 Integration Testing

Integration testing verifies that multiple EstateX components work together. A typical scenario begins with user authentication, continues to property discovery, opens a property detail view and finishes by submitting an enquiry. The purpose is to confirm that data flows correctly across frontend, backend and database layers.

Agent integration testing follows a different path: login, create a property, retrieve the listing, update a field and then delete the listing if required. The test confirms both data persistence and authorization behavior.

Integration testing also checks error paths. For example, if the backend is unavailable, the frontend should display an understandable status rather than silently failing. If a protected API receives an invalid token, the backend should reject the request and the frontend should handle the response appropriately.

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8.3 Tools Used for Testing

Postman is the primary API-testing tool for EstateX. Browser developer tools are useful for inspecting network requests, JSON responses, console errors and frontend behavior. MongoDB tools can be used to verify that records created through the API are actually stored as expected.

Manual browser testing is important for responsive UI behavior. The main workflows should be checked at desktop and mobile widths, including navigation, search, authentication, property details, listing forms and the EMI calculator.

Git also contributes to testing discipline because a known working commit can be restored if an experimental change introduces a regression. Automated tests can be added later as the codebase stabilizes.

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8.4 Test Cases and Results

Test cases for EstateX should cover both positive and negative scenarios. Example cases include retrieving the property list, applying a location filter, submitting a valid login, rejecting invalid credentials, creating a listing as an authorized agent, rejecting an unauthorized create request, and submitting a valid enquiry.

For each test, the expected result should be recorded alongside the actual result during demonstration. HTTP status codes such as 200 for successful retrieval, 201 for successful creation, 400 for validation errors and 401/403 for authentication or authorization failures can be used according to the implemented API contract.

Because this academic report is intended to describe the project rather than fabricate performance measurements, response-time or load figures should only be entered after the corresponding tests are actually executed. This keeps the documentation evidence-based.

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CHAPTER -9 DEPLOYMENT

9.1 Steps to Deploy

The local deployment process begins by installing Node.js and Python, creating the frontend and backend environments, installing dependencies and configuring the MongoDB connection. The frontend is then started with the Vite development command and the Flask backend is started on its configured port.

Before deployment, the production frontend URL and backend API URL should be configured through environment variables. The backend must receive its database URI and secret key from the hosting environment rather than from committed source code.

After building the React frontend, the generated static assets can be hosted on a suitable static web platform. The Flask application can be deployed to a server or managed application platform. MongoDB can remain local for academic demonstration or move to a managed deployment for public access.

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9.2 Environment Configuration

Environment configuration keeps deployment-specific values outside application source code. EstateX can use variables for the MongoDB URI, authentication secret, backend port and frontend API base URL. A local `.env` file is useful during development and should be excluded from Git commits.

The backend loads configuration at application startup and initializes the database connection. If the database is unavailable, the application should return a clear health or error status rather than presenting a misleading successful state.

Production configuration should use strong secrets, restricted CORS origins, HTTPS and appropriate database access controls. Separate development and production configurations reduce the risk of accidentally exposing development settings to public users.

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9.3 Hosting of Frontend and Backend

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For a production architecture, a managed MongoDB deployment can provide remote access and backups. The backend should be configured with the deployed database URI, and the frontend should point to the deployed API base URL rather than localhost.

Deployment should also include monitoring and logging. Health endpoints can be used to verify service availability, while application logs can help identify failed requests. The exact provider is a deployment choice and should be documented with the final hosting configuration used by the team.

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CHAPTER -10 CHALLENGES & LIMITATIONS

10.1 Issues Faced During Development

One challenge in EstateX development is coordinating the frontend and backend data contracts. The React application expects structured property fields, while the backend and database must return the same names and formats consistently. A mismatch can cause cards or filters to display incomplete information even when the database contains valid records.

Another challenge is configuring multiple local services. The frontend, Flask backend and MongoDB process use different ports and configuration values. Development is easier when these values are documented and health checks are available.

Authentication and authorization also require careful handling. A login interface may appear correct while a protected API still rejects requests if tokens are missing or incorrectly formatted. Testing the backend independently helps isolate such issues.

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10.1 Issues Faced During Development

One challenge in EstateX development is coordinating the frontend and backend data contracts. The React application expects structured property fields, while the backend and database must return the same names and formats consistently. A mismatch can cause cards or filters to display incomplete information even when the database contains valid records.

Another challenge is configuring multiple local services. The frontend, Flask backend and MongoDB process use different ports and configuration values. Development is easier when these values are documented and health checks are available.

Authentication and authorization also require careful handling. A login interface may appear correct while a protected API still rejects requests if tokens are missing or incorrectly formatted. Testing the backend independently helps isolate such issues.

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10.2 Solutions Applied

To reduce frontend/backend mismatch, the project uses a centralized API utility and a defined property data structure. Backend routes are organized by module, making it easier to locate and correct a specific endpoint without changing unrelated functionality.

Configuration issues are addressed through environment variables and clear local run commands. The backend exposes a health endpoint so that the frontend can determine whether the service is reachable. This is more reliable than assuming that the development server is always available.

Authentication problems are handled by separating login, token storage, authorization checks and protected operations. During testing, valid and invalid tokens are sent deliberately so that the backend behavior is understood before the feature is considered complete.

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10.3 Current Limitations or Known Bugs

The academic version of EstateX should be treated as a developing prototype rather than a fully commercial marketplace. Large-scale moderation, advanced fraud detection, payment processing, professional verification and enterprise-level analytics are not prerequisites for the current implementation.

Map-based discovery and 360-degree property viewing depend on the availability and quality of location and panoramic media. These features may require additional external services, assets and configuration before they can be treated as production-ready.

Performance at very high traffic levels has not been established through a production-scale load test. The documentation therefore avoids claiming specific concurrent-user capacity or response-time guarantees. Such measurements should be added only after controlled testing.

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CHAPTER -11 FUTURE ENHANCEMENTS

11.1 Planned Features

Future EstateX versions can introduce stronger property verification, document-based listing checks and an administrative approval workflow. Verification status can be visible on property cards so that users can distinguish between different listing states.

The platform can also introduce property comparison, saved searches and notifications. A user could save a search such as a rental property in a particular Hyderabad locality and receive an alert when matching inventory is added.

Mobile-first improvements and a dedicated mobile application are possible future directions. The same REST API can serve a mobile client, reducing the need to duplicate backend business logic.

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11.2 Possible Integrations

Map integration can evolve into a richer location experience with neighborhood information, nearby facilities and map-based search boundaries. A 360-degree viewer can be expanded with multiple rooms, navigation hotspots and property-tour metadata.

AI-assisted features can be added after reliable data collection is established. Examples include natural-language property search, recommendation based on preferences, lead prioritization and automated listing-quality checks. Such features should be introduced only after privacy, explainability and data-quality requirements are addressed.

Other possible integrations include cloud image storage, notification services, secure payment gateways, digital identity or document verification and analytics dashboards. Each integration should be isolated behind a stable service interface.

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CHAPTER -12 CONCLUSION

12.1 Summary of the Project

EstateX demonstrates the design of a full-stack real-estate portal that connects property discovery with agent-oriented listing and enquiry workflows. The project retains a clear separation between frontend presentation, backend services and database persistence.

The application focuses on practical real-estate tasks: searching by transaction type and location, reviewing structured property details, publishing listings, managing enquiries and using an EMI calculator. These functions are supported by a modular REST API and a document-oriented database.

The project also establishes a foundation for richer location and virtual-tour experiences. Because the architecture is modular, new features can be introduced without rebuilding the application from the beginning.

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EstateX is developed as a modular full-stack real-estate platform. The current section explains the corresponding design or implementation area and connects it with the broader workflow of property discovery, listing management and buyer-agent communication.

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For project demonstration, the most important evidence is the complete user flow. A reviewer should be able to see property search, listing details, authentication, agent listing management, enquiry submission and the supporting utility features. API requests and database records can then be shown separately to demonstrate that the visible interface is backed by an actual full-stack implementation rather than static screens.

12.2 What Was Achieved

EstateX is developed as a modular full-stack real-estate platform. The current section explains the corresponding design or implementation area and connects it with the broader workflow of property discovery, listing management and buyer-agent communication.

The design emphasizes maintainability, clear responsibilities and a predictable data flow between the React frontend, Flask backend and MongoDB database. Each module is documented so that future features can be added without unnecessarily changing unrelated components.

The implementation remains focused on an academic prototype while following practices that are useful for a future production system, including validation, authentication, testing, environment configuration and version control.

For future growth, the implementation should avoid hard-coding assumptions that only work for a small demonstration. Validation, modular routes and environment-based configuration make it easier to move from local development toward a hosted application without rewriting the core product.

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12.3 Skills Learned During Development

The team gained practical experience in component-based frontend development, responsive interface design and state management. Working with React also reinforced the importance of reusable components and clear data flow.

Backend development introduced REST API design, request validation, authentication, authorization, database access and environment configuration. MongoDB provided experience with document-oriented data modeling and querying.

The project also developed software-engineering skills such as debugging, Git version control, API testing, documentation and system architecture. These skills are directly transferable to larger full-stack and startup-oriented applications.

In the EstateX workflow, this component is connected to the wider property lifecycle: a user discovers a listing, examines its details, decides whether it fits the requirement, and then takes an action such as saving the property or contacting the agent. The software design therefore gives importance to clarity of information and predictable responses.

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The same principle applies to data consistency. A property shown in the UI should represent the record stored by the backend, and changes made by an authorized agent should become visible through subsequent API requests. This relationship between presentation, service logic and persistence is central to the project architecture.

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CHAPTER -13 REFERENCES

Framework and language documentation

React.js documentation — component-based user interfaces and application development. Vite documentation — frontend development and build tooling. Flask documentation — Python web application framework and routing. PyMongo documentation — Python access to MongoDB. MongoDB documentation — document database concepts, CRUD operations and indexing.

Web platform references

MDN Web Docs — HTML, CSS, JavaScript and browser APIs. W3C web standards — semantic web and accessibility guidance. Postman documentation — REST API testing, collections and environment variables. Git documentation — version control commands and workflows.

Project-specific references

EstateX project source code, API routes, database models, frontend components and project notes maintained by the development team. These materials form the primary implementation reference for the academic prototype.

CHAPTER -13 REFERENCES

Mapping and virtual-tour references

Mapbox documentation can be consulted for interactive maps, markers, geocoding and location-based interfaces when the EstateX map module is enabled. Documentation for the selected 360-degree viewer library or media format should be followed for panoramic property experiences.

Database and security references

MongoDB documentation for data modeling, indexes and query operations. Flask security guidance and Python cryptographic/password-hashing libraries should be followed for authentication. JWT-related documentation may be used where signed token authentication is implemented.

Academic and software-engineering references

General software-engineering texts and course material covering requirements analysis, system architecture, database design, REST APIs, testing and deployment were used to structure the report. External claims or performance figures should be verified against the actual EstateX implementation before final submission.

CHAPTER -14 APPENDICES

Screenshots of the App

This appendix section is reserved for screenshots of the EstateX application. The final submission should insert clear screenshots of the homepage, property-search interface, property details, login/register screens, agent listing form, enquiry flow and EMI calculator.

Screenshots should be captured from the actual running application so that the evidence matches the implementation described in this report. Each image should have a short caption identifying the feature and the user action demonstrated.



INSERT ESTATEX HOMEPAGE / SEARCH SCREENSHOT HERE

CHAPTER -14 APPENDICES

Sample Code Snippets

Representative code snippets may be inserted here from the EstateX frontend and backend. Suitable examples include the Flask application factory, a property API route, the React API utility, a property-card component and authentication logic.

Code included in the final academic report should be taken from the actual project repository and shortened to the relevant section so that the appendix remains readable.

INSERT VERIFIED CODE SCREENSHOT / SNIPPET HERE

CHAPTER -14 APPENDICES

Installation / Setup Instructions

Installation / Setup: Open the EstateX project in VS Code. Install frontend dependencies with npm install. Install the Python backend dependencies from the backend requirements file. Configure the MongoDB connection and authentication secret through environment variables.

Run the frontend: start the Vite development server and open the displayed localhost URL in a browser. **Run the backend:** start the Flask application on its configured port and verify the health endpoint. **Database:** ensure the local MongoDB service is running before testing database-backed features.

The exact commands used for the final version should be copied from the project README so that the appendix remains synchronized with the submitted source code.

CHAPTER -14 APPENDICES

User Manual / Guide

User Guide: Open EstateX and select Buy or Rent. Enter a locality or choose a property type, then review the available listings. Open a property to view detailed information, images and agent contact details. Use the enquiry action when interested in a property.

Agent Guide: Sign in using an authorized agent account, open the listing workflow, enter complete property information and publish the listing. Use the agent workflow to update or remove information when required.

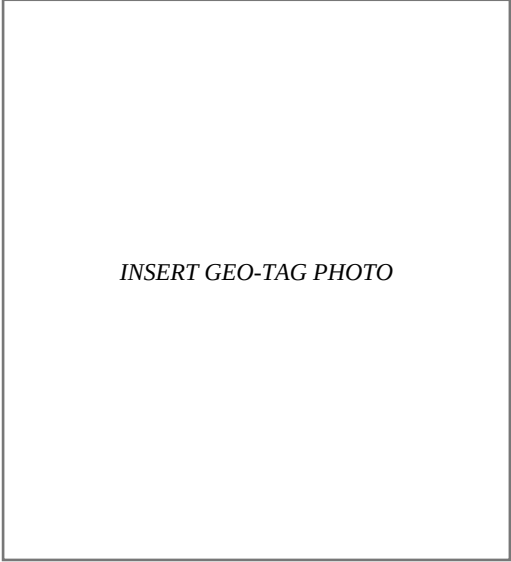
Utility Guide: Use the EMI calculator to estimate a periodic loan payment from the required financial inputs. Saved or favourite properties can be reviewed later where the corresponding module is enabled.

CHAPTER -14 APPENDICES

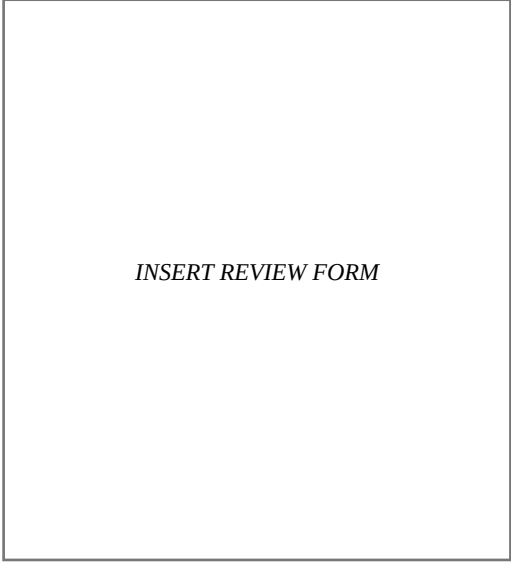
Geo-Tag Photos with Guide

Geo-Tag Photos with Guide: This section should contain the required project-review photographs with date/location metadata if such evidence is requested by the course. Only genuine project photographs should be inserted.

Review Evidence: Add photographs or scans of review forms, signatures and other required academic evidence after the guide/course coordinator completes them.



INSERT GEO-TAG PHOTO



INSERT REVIEW FORM

CHAPTER -14 APPENDICES

Review Forms with Guide Signatures

Final appendix checklist: verify that all screenshots represent the current EstateX build; confirm that code snippets match the GitHub repository; confirm that installation commands work on a clean development environment; attach required geo-tag photographs; and obtain the required guide/course-coordinator signatures.

The appendix is intentionally evidence-oriented. No fabricated screenshots, test results, photographs or signatures should be added. Replace each placeholder with material from the actual project before final academic submission.